

NEW PRODUCTS

COMPONENTS

DC/DC buck converter

Texas Instruments (TI) has introduced a wide-input-voltage (VIN) synchronous DC/DC buck converter with the combination of light load efficiency, ease of design and overall power solution cost. The 100V, 1A LM5164 step-down voltage converter shrinks board space in rugged battery-powered industrial and automotive power supplies. Used in conjunction with TI's WEBENCH power designer, this new DC/DC converter can enable simpler, faster power conversion design.

Texas Instruments Inc.
www.ti.com

LED controller ICs

With BCR601 and BCR602 Infineon Technologies introduces two new members of its successful BCR linear LED controller IC portfolio. BCR601 features an innovative voltage feedback to the primary side, also known as active headroom control (AHC), enabling cost- and power-effective LED driver applications. BCR602, on the other hand, targets dimmable LED applications such as light engines, modules and strips. Its wide voltage range of up to 60V makes it an ideal fit for 48V designs and DC/DC grids.

Infineon Technologies India Pvt Ltd
www.infineon.com

Power connector

Samtec announces the release of mPOWER connectors, a micro high-power solution with design flexibility for power-only or power/signal applications. This 2.00mm pitch power connector system

(UMPT/UMPS) features a small form factor, which saves board space, while achieving up to 18A per blade.

Samtec Inc.
www.samtec.com

SiC MOSFET

ON Semiconductor introduces two silicon-carbide (SiC) MOSFET devices. The industrial grade NTHL080N120SC1 and AEC-Q101 automotive grade NVHL080N120SC1 bring the enabling, wide-ranging performance benefits of wide-band gap technology to important high-growth end application areas such as automotive DC-DC and onboard charger applications for electric vehicles (EVs) as well as solar, and uninterruptible and server power supplies.

ON Semiconductor
www.onsemi.com

USB-C buck charger

Designers of portable lithium-ion battery-powered electronics now have a simplified and more flexible way to add a USB Type-C charging system to their products with MAX77860 3A switch-mode charger from Maxim Integrated Products. This USB Type C buck charger eliminates the need for a separate host controller, simplifies software development and reduces overall bill-of-materials (BOM) costs for applications such as financial point-of-sale terminals, power banks, industrial computers, scanners, radios, medical devices, charging cradles, portable speakers and game players.

Maxim Integrated Products Inc.
www.maximintegrated.com

Flash MCU

Renesas Electronics has announced the world's first microcontroller (MCU)

with embedded flash that integrates a hardware-based virtualisation-assisted function while maintaining the fast, real-time performance of RH850



products. This hardware-based virtualisation assist technology can support up to ASIL D level of functional safety, providing greater levels of system integration. The RH850/U2A MCU is the first member of Renesas' cross-domain MCUs, a new generation of automotive-control devices designed to address the growing need to integrate multiple applications into a single chip to realise a unified electronic control unit for the evolving electrical-electronic architecture.

Renesas Electronics Corp.
www.renesas.com

Operational amplifier

ADHV4702-1 is a high voltage (220V), unity-gain stable precision operational amplifier. It offers high input impedance with low input bias current, low input offset voltage, low drift and low noise for precision demanding applications. Key features are:

- Wide range of operating voltages
- Dual supply: $\pm 12V$ to $\pm 110V$
- Asymmetrical supply operation: 24V to 220V
- Wide input common-mode voltage range: 3V from rails
- High common-mode rejection ratio: 160dB typical
- High AOL: 170dB typical
- Output voltage: 0.6V to 5.5V
- High slew rate: 74V/ μs typical; 24V/ μs typical with external input clamping diodes

Analog Devices
www.analog.com

32-bit MCUs

Holtek HT32F52344/52354 devices are high-performance, low power consumption 32-bit MCUs based around Arm Cortex-M0+ processor core. The